

Sparse binary matrices with line sums at most two

Exact counts and all fixed order asymptotics

Report 168

3 October 2026

Abstract

Let a_n count $n \times n$ binary matrices with at most two ones in every row and column, with $a_0 = 1$. This is OEIS A197458. We prove

$$a_n \sim \frac{(n!)^2}{2^{5/4} \pi n^{3/4}} e^{2\sqrt{2n}},$$

and an expansion to every fixed order in $n^{-1/2}$, with four explicit corrections and a finite exact Gaussian-moment rule for all subsequent coefficients. The proof starts from an elementary labeled-component decomposition and a modified-Bessel diagonal. A sectorial Bessel remainder, a whole-circle bound, explicit off-circle connector estimates, and an integrable Taylor remainder give the claimed sharp fixed-order error.

We also derive a logarithmic expansion, an explicit Lambert- W real-inverse formula through its bounded correction, and a rigorous two-ceiling enclosure for the integer threshold. This enclosure retains the distinction between a smooth comparison inverse and the actual discrete crossing. The companion supplies bounded exact counts, independent finite checks, and optional symbolic coefficient checks.

General bounded-degree enumeration and configuration-model methods are classical. This is an explicit specialization and refinement; no historical-priority assertion is made. The results are fixed-order asymptotics, not a complete beyond-all-orders description. Remainder constants and eventual thresholds are existential, not numerically certified.

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 168 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a197458-line-sum-two-matrices** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 168” and its PDF author field is empty. *What rests on what.* The generating function (2.1), the Bessel diagonal (2.3), the differential equation (3.1) and the recurrence (3.3) are exact identities; the first is equivalent to a formula recorded in OEIS A062154 (Remark 2.1). Theorem 1.1, Corollary 1.2 and Theorems 7.1 and 7.2 are proved by the source’s own saddle-point argument (Sections 4–7), which uses no computation and no external asymptotic theorem beyond Stirling’s formula; their constants and onsets are existential. The ODE route to $c_1, \dots, c_4$ after the proof of Theorem 1.1 is an identity check only. The write adds, with proofs: Remark 2.1 (the OEIS entries A197458, A062154, A062156 and A001499, quoted; the source’s generating function as e^{x+y-xy} times the A062154 formula; the exactly-two family A001499 inside the same decomposition, with Knuth’s equivalent and the ratio $a_n/A001499(n)$); Remark 7.3 (the inverses against the transseries volume: N is an instance of the factorial core, the expansion (7.4) truncates an instance of the formal core reversion, and Theorem 7.2 follows from the staircase theorem through an explicit admissible interpolation); Lemma 9.1 and Proposition 9.2 (the expansion for $(n+d) \times n$ matrices with d fixed, and an independent Poisson(1) limit for the numbers of zero rows and zero columns, which advance two of the source’s questions); notes on the reruns and on the literature; and the numbered open questions at the end of Section 9. Nothing was submitted to the OEIS.

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Reading guide. Section 1 states the asymptotic results. Sections 2–6 prove the exact formulas and the coefficient expansion. Section 7 proves real and integer inversion. Sections 8–10 record reproducibility and the precisely bounded source coverage. All analytic arguments used for the main theorem are included.

1 Results and conventions

Rows and columns are individually labeled. A line means either a row or a column. Write $b_n = a_n/(n!)^2$, and use $[z^k]$ for coefficient extraction. Every asymptotic statement below is as its argument tends to positive infinity. An order index such as J is fixed before the limit; implied constants may depend on it.

Theorem 1.1. *For each integer $J \geq 0$ there are constants $c_j \in \mathbb{Q}(\sqrt{2})$ such that*

$$a_n = \frac{(n!)^2 e^{2\sqrt{2}n}}{2^{5/4} \pi n^{3/4}} \left(\sum_{j=0}^J c_j n^{-j/2} + O_J(n^{-(J+1)/2}) \right). \quad (1.1)$$

The first five coefficients are

$$\begin{aligned} c_0 &= 1, & c_1 &= -\frac{17\sqrt{2}}{96}, & c_2 &= \frac{6409}{9216}, \\ c_3 &= -\frac{18114133\sqrt{2}}{13271040}, & c_4 &= \frac{13052721749}{2548039680}. \end{aligned} \quad (1.2)$$

The finite rule in Section 6 specifies every c_j .

The theorem asserts a Poincaré expansion at every fixed order. It neither asserts convergence of the formal coefficient series nor identifies every exponentially small contribution. In particular, letting J grow with n is not justified by the theorem.

Corollary 1.2. *The logarithmic expansion begins*

$$\begin{aligned} \log a_n &= 2n(\log n - 1) + 2\sqrt{2}n + \frac{1}{4} \log n - \frac{1}{4} \log 2 \\ &\quad - \frac{17\sqrt{2}}{96\sqrt{n}} + \frac{319}{384n} + O(n^{-3/2}). \end{aligned} \quad (1.3)$$

There is a logarithmic expansion to every fixed half-integer order.

For the latter assertion, formally expand $\log(\sum c_j n^{-j/2})$ to the desired finite order and add twice Stirling's logarithmic series. The constant and first correction check is

$$c_2 - \frac{1}{2}c_1^2 = \frac{85}{128}, \quad \frac{85}{128} + \frac{1}{6} = \frac{319}{384}.$$

Positivity of the leading normalized quantity for sufficiently large n justifies taking its logarithm with the stated remainder.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 108.** *Provenance.* This report is Report 168 of the session bundle that ProveIt's docs/incoming intake processed as batch 108: the archive `Sparse_Binary_Matrices_All_Orders_and_Inverses_Source.zip` (523,157 bytes, SHA-256 `b67e3c4e515b...32a02a9a0b8`, 24 files at the archive root, no wrapper directory), arrival commit `60f54ea06`, placed in `602e5bd0f`. The text is the delivered `Report168.tex` (879 lines, dated 3 October 2026), shipped as `article.tex`. The delivered 15-page PDF and the checksum manifest `SHA256SUMS` (23 entries, verified at placement) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival

commit. The package records no ProveIt commit (it cites nothing in the repository), no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 50 delivered labels with `sbm:` and updated the 45 references to them; it added this subsection, the notes marked [write], Remarks 2.1 and 7.3, Lemma 9.1, Proposition 9.2, the numbered questions at the end of Section 9 and two bibliography entries. No statement, proof or number of the source was changed: the added statements are the last numbered items of their sections, the added subsection is the only numbered subsection, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A197458 [1]: the live entry is still revision #25 (8 April 2020), the revision the source used; its 17 terms are frozen in `data/oeis_fixtures.json`, and it still has no asymptotic formula. Its name is quoted in Remark 2.1; the source’s bibliography paraphrases it.
- OEIS A062154 (revision #14, 18 August 2024), A062156 (#7, 3 February 2021) [4] and A001499 (#130, 21 September 2026) [5], quoted in Remark 2.1; A283500 and A247158 [8], which list A197458 as the column $k = 2$ of the square “at most k ones” family and cite Mathar’s note.
- The abstract of Ordentlich, Parvaresh and Roth [9] (through Crossref), on the regime $k = n/2$ of that family; the source does not cite it.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:prop:factorial-core`, `p0:rem:factorial-core`, `p0:thm:core-reversion`, `p0:rem:core-instances`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 7.3.
- Not read by the write: Mathar’s note [2], the Math Stack Exchange thread [3], Goulden and Jackson’s exercise, Knuth’s solution, Greenhill, McKay and Wang [6], and Caizergues and de Panafieu [7]. What Section 10 says about them is the source’s; `data/SOURCE_PROVENANCE.json` records what the source inspected.

What was checked at intake. The batch-108 dossier read the manuscript, recomputed the constants of Corollary 1.2, and reran the companion on copies (all 17 OEIS terms, the ODE recurrence to $n = 200$, the four corrections; 35 companion tests pass; 10 of the 11 release tests error on Windows, which the delivered README says need POSIX). The write rechecked the proofs line by line and found no error; it rederived the component table and (2.1), the loss (5.1), the endpoint difference (5.6), the scaling (6.1) and the leading constant, the Bessel coefficients γ_k , the expansions (4.6) and the inverse residual (7.6) with δ_1 , and made the independent computations recorded at the end of Section 8 (terms of A197458, A062156 and A001499 by its own programs; Theorem 1.1, Corollary 1.2 and Theorem 7.1 against exact values to $n = 4000$). Finite computations are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository mentioned A197458, A062154, A062156, A001499 or binary matrices with line sums at most two, and none does outside this report and the collection catalogue. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The inverses of Section 7 are compared with the transseries volume in Remark 7.3.

[write] What is not claimed, collected from the source. No historical priority is claimed: general bounded-degree enumeration and configuration-model methods are classical, the component generating functions are older (Remark 2.1 makes this precise), and the literature check is bounded (the Goulden–Jackson exercise and Knuth’s solution were not inspected). Theorem 1.1 is a Poincaré expansion at each fixed order: it asserts no convergence of the coefficient series, does not identify the exponentially small contributions or give a transseries, and does not justify letting J grow with n . No remainder constant or onset is effective; Theorem 7.2 has existential constants and is not a certified finite-input rounding algorithm, and no universally correct rounding of an asymptotic inverse is asserted. The ODE route to $c_1, \dots, c_4$ proves neither the existence of the expansion nor its leading constant. Finite exact agreement checks implementations and identities, not the asymptotic theorem, and decimal diagnostics are uncertified. The four directions of Section 9 are not proved there. The write adds: Remark 7.3 compares statements and proves no new inverse;

its part (c) re-derives Theorem 7.2 and claims nothing stronger; Proposition 9.2 covers bounded offsets $m - n$ and the zero lines only, with existential constants; the decimal values printed at the ends of Sections 8 and 9 are floating-point observations.

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol of the source was renamed; F_{154} , τ , ξ , h_{vol} , f_{vol} , d , ω_1 , ω_2 , Z_r , Z_c are the write's).

Symbol	Meaning here (where)	Not to be read as
$a_n, b_n, A_{m,n}$	A197458; $a_n/(n!)^2$; the rectangular count (Section 2)	the A_n of <code>p0:df:three-inverses</code> (which is a_n in Remark 7.3)
F, F_J, F_{154}	the bivariate EGF (2.1); the smooth comparison (7.1); the A062154 EGF (Remark 2.1)	one function
$q(t), q$	the Bessel argument $2\sqrt{t}h(t)$ of (2.2) (a complex variable in Lemma 4.1); $q = \log N$ in Theorem 7.1	each other: the source uses q for both
$h(t), h_j, H$	$(2-t)/(2(1-t))$; the coefficients of the amplitude $H(s)$ of (4.5)	h_{vol} of <code>p0:thm:core-reversion</code>
$L(t), L$	P'/P (Section 3); $\log y$ (Section 7)	each other
$I_0, I(y)$	the modified Bessel function; the integer crossing of Theorem 7.2	each other
$p(s), p_i$	$\sqrt{1-s}(1+s)$ of (4.3); the ODE polynomials (3.2)	the displacement E of <code>p0:thm:core-reversion</code> (here ξ)
$E(s), E_m$	the real phase of Section 5; the coefficients of e^Ψ in (6.5)	
N, n	$L/(2W(L/(2e)))$, a real number; the matrix size	the integer crossing $I(y)$, or the staircase N_* of the volume
W	the branch W_0 on $(0, \infty)$	W_{-1}
$\varepsilon, u, \mathcal{M}, M$	$n^{-1/4}$; the Gaussian variable; the moment functional (6.3); a truncation order	u of <code>p0:thm:core-reversion</code>
$D(r, \theta), D_J; \delta, \delta_0, \delta_1$	the loss (5.1), an exponent in (6.6); the sector margin of Lemma 4.1, the inverse corrections	Euler's number e
$e(t), \alpha, z$	$t^{-\alpha}/\log t$ with $\alpha = (J+1)/2$ (proof of Theorem 7.2); $z = n^{-1/2}$ (the ODE check)	
$d, c_j^{(d)}, T_d$	a fixed offset $m - n$ (Proposition 9.2); its coefficients; the Chebyshev polynomial	the volume's d in Remark 7.3(a)

2 Exact component decomposition and diagonal

A matrix is a simple bipartite graph with specified shores of size n . Maximum degree two means that each connected component is an isolated vertex, a path, or an even cycle. Let $A_{m,n}$ count the rectangular class and define its bivariate exponential generating function

$$F(x, y) = \sum_{m, n \geq 0} A_{m, n} \frac{x^m y^n}{m! n!}.$$

For $t = xy$, the connected component contributions are as follows:

Component	Contribution
Isolated vertices	$x + y$
Balanced paths on $k + k$ vertices, $k \geq 1$	t^k
Unbalanced paths on $(k + 1) + k$ vertices, $k \geq 1$	$(x + y)t^k/2$
Even cycles on $k + k$ vertices, $k \geq 2$	$t^k/(2k)$

A balanced path has $(k!)^2$ realizations on specified labels: choose its ordering starting from its unique endpoint in the first shore. An unbalanced path has a reversal symmetry of order two. A cycle has $2k$ equivalent oriented starting descriptions. The lower cutoff $k = 2$ for cycles excludes the parallel-edge component forbidden in a binary matrix. The exponential formula for sets of labeled connected components now gives

$$F(x, y) = (1 - t)^{-1/2} \exp \left\{ \frac{t}{1 - t} - \frac{t}{2} + (x + y) \frac{2 - t}{2(1 - t)} \right\}. \quad (2.1)$$

This identity also follows directly by multiplying the exponential series for the four kinds of components, so no asymptotic input is being invoked.

Put

$$P(t) = (1 - t)^{-1/2} e^{t/(1-t)-t/2}, \quad h(t) = \frac{2 - t}{2(1 - t)}, \quad q(t) = 2\sqrt{t} h(t). \quad (2.2)$$

The diagonal requires equal powers from $e^{xh(t)}$ and $e^{yh(t)}$. Consequently

$$G(t) := \sum_{n \geq 0} b_n t^n = P(t) \sum_{k \geq 0} \frac{t^k h(t)^{2k}}{(k!)^2} = P(t) I_0 \left(\frac{\sqrt{t}(2 - t)}{1 - t} \right), \quad (2.3)$$

where

$$I_0(z) = \sum_{k \geq 0} \frac{(z^2/4)^k}{(k!)^2}.$$

The square root is harmless because I_0 is even. In particular, $q(t)^2 = t(2 - t)^2/(1 - t)^2$ is rational, and G is analytic in $|t| < 1$ with nonnegative coefficients.

Remark 2.1 ([write] The four OEIS entries, quoted, and the older generating functions, 6 October 2026). The entries were read on 6 October 2026; their names, verbatim, are: A197458, “Number of $n \times n$ binary matrices with at most two 1’s in each row and column, other entries 0.” (Felix A. Pahl, 2011; revision #25, 8 April 2020); A062154, “Number $T(n, m)$ of $n \times m$ matrices over $\{0, 1, 2\}$ with all row and column sums equal to 1 or 2, $m=0, \dots, 2^*n$.” (Vladeta Jovovic, 2001; revision #14); A062156, “Number of $n \times n$ matrices over $\{0, 1, 2\}$ with all row and column sums equal to 1 or 2.” (its main diagonal); and A001499, “Number of $n \times n$ matrices with exactly 2 1’s in each row and column, other entries 0.” (revision #130). The source’s account of them in Section 10 is accurate; the following makes “closely related” precise.

- (a) *The generating function (2.1) is the A062154 formula times e^{x+y-xy} .* The A062154 entry gives (typeset from its plain-text formula line; the README quotes the line verbatim)

$$F_{154}(x, y) := \sum_{n, m \geq 0} T(n, m) \frac{x^n y^m}{n! m!} = \frac{1}{\sqrt{1 - xy}} \exp \left(\frac{xy}{2} + \frac{xy + x^2 y/2 + xy^2/2}{1 - xy} \right).$$

With $t = xy$ its exponent is $\frac{t}{2} + \frac{t}{1-t} + (x + y) \frac{t}{2(1-t)}$, while that of (2.1) is $\frac{t}{1-t} - \frac{t}{2} + (x + y) \left(1 + \frac{t}{2(1-t)}\right)$; the difference is $x + y - t$, so

$$F(x, y) = e^{x+y-xy} F_{154}(x, y).$$

Combinatorially, e^{x+y} adjoins zero rows and columns, and e^{-xy} removes the components that consist of a single entry 2 (such an entry fills its row and its column, so it is a whole component, with exponential generating function xy); what remains are exactly the binary matrices with

line sums at most two. So (2.1) follows in one line from a formula of the OEIS entry, which dates from June 2001 (the formula line is present in the earliest snapshot of the OEIS data repository, February 2023; when it was added was not traced). The entry's only literature reference is Goulden and Jackson's Problem 3.4.15 (not inspected by the source or the write). The source derives (2.1) independently and claims no priority for it. By the extraction of (2.3), A062156 also has a Bessel diagonal: $\sum_n A062156(n) t^n / (n!)^2 = (1-t)^{-1/2} e^{t/2+t/(1-t)} I_0(t^{3/2}/(1-t))$.

- (b) *The exactly-two family.* The components all of whose vertices have degree two are the even cycles of the table in Section 2 ($k \geq 2$), so $B(t) := \sum_n A001499(n) t^n / (n!)^2 = (1-t)^{-1/2} e^{-t/2}$, the formula the A001499 entry attributes to Bricard ("Sum_{n >= 0} a(n)*x^n/(n!)^2 = exp(-x/2)/sqrt(1-x)"). Hence $P(t) = B(t) e^{t/(1-t)}$ and $G(t) = B(t) e^{t/(1-t)} I_0(q(t))$. From B alone, $A001499(n)/(n!)^2 \sim e^{-1/2}/\sqrt{\pi n}$, which with Stirling's formula is the equivalent the entry attributes to Knuth, "a(n) ~ 2 sqrt(Pi) n^(2n + 1/2) e^(-2n - 1/2)". With Theorem 1.1 it follows that

$$\frac{a_n}{A001499(n)} \sim \frac{e^{1/2} e^{2\sqrt{2n}}}{2^{5/4} \sqrt{\pi} n^{1/4}}.$$

Proof of the equivalent. Let $u_m = \binom{2m}{m} 4^{-m}$, the coefficients of $(1-t)^{-1/2}$. Then $u_m^2 = \prod_{i=1}^m \left(\frac{2i-1}{2i}\right)^2 \leq \prod_{i=1}^m \frac{2i-1}{2i} \cdot \frac{2i}{2i+1} = \frac{1}{2m+1}$, and $u_m \sqrt{\pi m} \rightarrow 1$ (Wallis). Now $\sqrt{n} [t^n] B(t) = \sum_{k=0}^n \frac{(-1/2)^k}{k!} \sqrt{n} u_{n-k}$. For $k \leq n/2$ the k -th term is at most $2^{-k}/k!$ in absolute value, since $\sqrt{n} u_{n-k} \leq \sqrt{n/(2(n-k)+1)} \leq 1$, and it tends to $\frac{(-1/2)^k}{k!} \pi^{-1/2}$ for each fixed k ; the terms with $k > n/2$ add up to at most $\sqrt{n} \sum_{k>n/2} 2^{-k}/k! \rightarrow 0$. By dominated convergence $\sqrt{n} [t^n] B(t) \rightarrow e^{-1/2} \pi^{-1/2}$. Stirling's formula $(n!)^2 \sim 2\pi n^{2n+1} e^{-2n}$ gives Knuth's form, and dividing the equivalent of Theorem 1.1 by $e^{-1/2}/\sqrt{\pi n}$ gives the ratio.

Nothing was submitted to the OEIS.

3 Differential equation and exact recurrence

Let

$$R(t) = \frac{t(2-t)^2}{(1-t)^2}, \quad L(t) = \frac{P'(t)}{P(t)} = \frac{1}{2(1-t)} + \frac{1}{(1-t)^2} - \frac{1}{2}.$$

Termwise differentiation shows that $f(R) = I_0(\sqrt{R})$ solves $Rf'' + f' - f/4 = 0$. Under the pullback $R = R(t)$ define

$$A = \frac{R'}{R} - \frac{R''}{R'}, \quad B = \frac{R'^2}{4R}.$$

The pulled-back solution satisfies $Y'' + AY' - BY = 0$. Conjugating by P , then clearing denominators, yields

$$p_2(t)G'' + p_1(t)G' + p_0(t)G = 0, \tag{3.1}$$

with

$$\begin{aligned} p_0(t) &= t^6 - 2t^5 - 3t^4 + 20t^3 - 36t^2 + 32, \\ p_1(t) &= 4t^6 - 12t^5 + 8t^4 + 28t^3 - 92t^2 + 96t - 16, \\ p_2(t) &= 4t^6 - 20t^5 + 44t^4 - 60t^3 + 48t^2 - 16t. \end{aligned} \tag{3.2}$$

For example, the unnormalized coefficients before clearing denominators are $A - 2L$ and $L^2 - L' - AL - B$; this gives a short rational identity check of every displayed polynomial.

Write $p_{i,k} = [t^k]p_i(t)$ and interpret negative-index b 's as zero. The coefficient equation is

$$\begin{aligned} 0 = & \sum_{k=0}^6 p_{2,k}(n-k+2)(n-k+1)b_{n-k+2} \\ & + \sum_{k=0}^6 p_{1,k}(n-k+1)b_{n-k+1} + \sum_{k=0}^6 p_{0,k}b_{n-k}. \end{aligned} \quad (3.3)$$

The coefficient of b_{n+1} is $-16(n+1)^2$. Thus $b_0 = 1$ uniquely determines the analytic power series. For a recurrence involving only integers, multiply the other terms by $(n!)^2$ before solving for a_{n+1} : since $16(n+1)^2 b_{n+1}(n!)^2 = 16a_{n+1}$, every remaining term is an integer multiple of $a_m(n!/m!)^2$ with $0 \leq m \leq n$. Division by 16 is exact, as follows either from the generating function or the combinatorial interpretation. The companion checks this division at every step rather than silently truncating it.

4 A sectorial Bessel expansion with remainder

The following elementary proof specifies the uniformity needed later. For all complex q ,

$$I_0(q) = \frac{1}{\pi} \int_0^\pi e^{q \cos v} dv = \frac{e^q}{\pi} \int_0^2 e^{-qw^2/2} (1 - w^2/4)^{-1/2} dw. \quad (4.1)$$

The first identity follows by expanding the exponential and integrating powers of cosine; the second uses the real substitution $w = 2 \sin(v/2)$.

Lemma 4.1. *For any fixed $M \geq 1$ and $\delta > 0$, uniformly as $|q| \rightarrow \infty$ in $|\arg q| \leq \pi/2 - \delta$,*

$$I_0(q) = \frac{e^q}{\sqrt{2\pi q}} \left(\sum_{k=0}^{M-1} \gamma_k q^{-k} + O_{M,\delta}(|q|^{-M}) \right), \quad \gamma_k = \frac{(1 \cdot 3 \cdots (2k-1))^2}{k! 8^k}, \quad (4.2)$$

where $\gamma_0 = 1$ and the square root is positive on the positive axis.

Proof. On $0 \leq w \leq 1$ expand

$$(1 - w^2/4)^{-1/2} = \sum_{k=0}^{M-1} \frac{\binom{2k}{k}}{16^k} w^{2k} + O_M(w^{2M}).$$

Since $\operatorname{Re} q \geq |q| \sin \delta$, the integral of the remainder is $O_{M,\delta}(|q|^{-M-1/2})$. The interval $1 \leq w \leq 2$ contributes an exponentially small quantity: the endpoint singularity is integrable and $|e^{-qw^2/2}| \leq e^{-|q| \sin \delta/2}$ there. Extending each retained monomial integral from $[0, 1]$ to $[0, \infty)$ has the same type of exponentially small error. Finally,

$$\int_0^\infty e^{-qw^2/2} w^{2k} dw = 2^{k-1/2} \Gamma(k + \frac{1}{2}) q^{-k-1/2}.$$

This is a real substitution for positive q , and then holds throughout $\operatorname{Re} q > 0$ by holomorphic continuation; local domination makes the integral holomorphic there. Combining factors proves the formula. No complex substitution of the original real integration path is used. $\square$

Set $s = 1 - t$ and

$$p(s) = \sqrt{1-s}(1+s), \quad q(t) = p(s)/s, \quad (4.3)$$

where p is analytic and positive at zero. Direct substitution into (2.3) gives, in a sufficiently narrow fixed sector about positive real s ,

$$G(1-s) = \frac{e^{-1}}{\sqrt{2\pi}} e^{2/s} \left(\sum_{j=0}^{M-1} h_j s^j + O_M(|s|^M) \right). \quad (4.4)$$

The formal series $H(s) = \sum h_j s^j$ is specified by

$$H(s) = p(s)^{-1/2} \exp \left(\frac{p(s)-1}{s} - \frac{1}{2} + \frac{s}{2} \right) \sum_{k \geq 0} \gamma_k \left(\frac{s}{p(s)} \right)^k. \quad (4.5)$$

Only finite truncations are used. The first factors are analytic at zero: the apparent singularity in $(p(s)-1)/s$ is removable. Their product with a finite Bessel expansion gives the stated remainder, but no convergence of the infinite formal H is asserted. In particular,

$$\begin{aligned} H(s) &= 1 - \frac{s}{4} + \frac{7s^2}{32} - \frac{47s^3}{384} + \frac{1361s^4}{6144} + O(s^5), \\ \frac{p(s)-1}{s} - \frac{1}{2} + \frac{s}{2} &= -\frac{s}{8} - \frac{3s^2}{16} - \frac{13s^3}{128} - \frac{17s^4}{256} + O(s^5). \end{aligned} \quad (4.6)$$

Here and subsequently a finite-order O attached to a formal series means the corresponding finite truncation, justified analytically by (4.4) when it is used in a contour integral.

5 Whole-circle control and contour deformation

Choose

$$s_0 = \sqrt{2/n}, \quad r = 1 - s_0, \quad t = r e^{i\theta}, \quad -\pi \leq \theta \leq \pi.$$

We use only sufficiently large n , so $0 < r < 1$. Because $q(t) = \sqrt{t}(1 + (1-t)^{-1})$,

$$|q(t)| \leq \sqrt{r}(1 + (1-r)^{-1}) = q(r).$$

The nonnegative coefficients of I_0 give $|I_0(q(t))| \leq I_0(|q(t)|) \leq I_0(q(r))$. Also $|1-t| \geq 1-r$. Define the exact loss

$$D(r, \theta) = \frac{r}{1-r} - \operatorname{Re} \frac{t}{1-t} = \frac{r(1+r)(1-\cos \theta)}{(1-r)|1-r e^{i\theta}|^2}. \quad (5.1)$$

The remaining factor $e^{-t/2}$ costs at most e , and hence

$$|G(r e^{i\theta})| \leq e G(r) e^{-D(r, \theta)}. \quad (5.2)$$

Uniformly on the whole circle,

$$D(r, \theta) \geq c \min\{\theta^2/s_0^3, 1/s_0\} \quad (5.3)$$

for an absolute $c > 0$. Indeed $1 - \cos \theta$ is bounded above and below by positive multiples of θ^2 on $[-\pi, \pi]$, and $|1 - r e^{i\theta}|^2 = s_0^2 + 2r(1 - \cos \theta)$.

Take $\theta_0 = s_0^{3/2} \log n$. By (5.2) and (5.3), the coefficient integral outside $|\theta| \leq \theta_0$ is bounded by

$$O\left(r^{-n} G(r) e^{-c \log^2 n}\right). \quad (5.4)$$

The positive-axis instance of (4.4) shows $r^{-n}G(r) = O(e^{2\sqrt{2n}})$. Thus (5.4) is smaller than $e^{2\sqrt{2n}}n^{-3/4}$ by every fixed power of n . This estimates all other Bessel sectors without using the local Bessel expansion there.

On the retained arc, $s = 1 - t$ satisfies $|s| \asymp s_0$ and $\arg s = O(n^{-1/4} \log n)$, so (4.4) is uniform. To see explicitly that its remainder retains the Gaussian width, observe the exact circle identity

$$\operatorname{Re}(2/s) = 2/s_0 - 2D(r, \theta). \quad (5.5)$$

An $O(s_0^M)$ amplitude remainder contributes at most a constant times

$$s_0^M r^{-n} e^{2/s_0} \int_{-\theta_0}^{\theta_0} e^{-2D(r, \theta)} d\theta = O\left(s_0^{M+3/2} r^{-n} e^{2/s_0}\right).$$

The equality follows from (5.3) and the substitution $\theta = s_0^{3/2} v$. Relative to the saddle scale this is $O(s_0^M)$, without any extra power or logarithm of n .

For a particularly simple coefficient calculator we deform the local arc into the vertical segment

$$s = s_0 + iv, \quad |v| \leq v_n := r \sin \theta_0.$$

At each end, a horizontal connector joins this segment to the original arc, whose real coordinate is $1 - r \cos \theta_0$. Connector length is $O(\theta_0^2)$. The region is analytic and avoids $s = 0$, and stays inside $|t| < 1$ for large n . Define the real phase

$$E(s) = \operatorname{Re}\{2/s - (n+1) \log(1-s)\}.$$

At a vertical endpoint the exact difference is

$$E(s_0 + iv_n) - E(s_0) = -\frac{2v_n^2}{s_0(s_0^2 + v_n^2)} - \frac{n+1}{2} \log(1 + v_n^2/r^2) \leq -c \log^2 n. \quad (5.6)$$

On either connector,

$$\left| \frac{\partial E}{\partial \operatorname{Re} s} \right| \leq \frac{2}{|s|^2} + \frac{n+1}{|1-s|} = O(n).$$

Therefore its phase varies by only $O(n\theta_0^2) = O(n^{-1/2} \log^2 n) = o(1)$. The finite amplitude from (4.4) is bounded there. The connector integrals are again smaller than the saddle term by every power. This separate estimate is essential: the circle-only bound (5.2) was not applied to an off-circle path.

6 Gaussian coefficient rule and sharp remainder

Let $\varepsilon = n^{-1/4}$ and put

$$s = \sqrt{2} \varepsilon^2 + i 2^{1/4} \varepsilon^3 u. \quad (6.1)$$

The vertical endpoints satisfy $|u| \sim \sqrt{2} \log n$. Define

$$\Psi(\varepsilon, u) = \frac{2}{s} - (\varepsilon^{-4} + 1) \log(1-s) - 2\sqrt{2} \varepsilon^{-2} - 1 + u^2. \quad (6.2)$$

Expanding the rational term and logarithm shows that the negative powers cancel, and the constant term is zero. Thus $\Psi \in \varepsilon \mathbb{C}[u][[\varepsilon]]$. Notice the $n+1$ in this expression, which comes from the coefficient kernel $(1-s)^{-n-1}$.

Define the Gaussian functional $\mathcal{M}$ on polynomials by

$$\mathcal{M}(u^{2k}) = \frac{(2k)!}{4^k k!}, \quad \mathcal{M}(u^{2k+1}) = 0. \quad (6.3)$$

Equivalently, $\mathcal{M}(Q) = \pi^{-1/2} \int_{\mathbb{R}} e^{-u^2} Q(u) du$. The promised all-fixed-order rule is

$$c_j = \mathcal{M} \left([\varepsilon^{2j}] H(s) e^{\Psi(\varepsilon, u)} \right). \quad (6.4)$$

For a requested j , retain H through degree j , expand Ψ through degree $2j$, and perform finite polynomial multiplication. Although $2^{1/4}$ and i appear in the intermediate scaling, only even powers survive the Gaussian moments, giving $c_j \in \mathbb{Q}(\sqrt{2})$. One can also see this from the parity of the powers of u in the expansion. All operations are exact algebraic operations.

For an explicit implementation, if $\Psi = \sum_{m \geq 1} \psi_m(u) \varepsilon^m$ and $e^\Psi = \sum E_m(u) \varepsilon^m$, use

$$E_0 = 1, \quad E_m = \frac{1}{m} \sum_{k=1}^m k \psi_k E_{m-k}. \quad (6.5)$$

It follows by differentiating the formal exponential. Equations (4.5), (6.2), (6.3), and (6.5) are a finite calculator at every requested fixed order, independent of any numerical fitting.

Proof of Theorem 1.1. The discarded circle arcs and horizontal connectors are already controlled. On the vertical segment substitute (4.4) and (6.1) into Cauchy's formula. The leading constant is

$$\frac{e^{-1}}{\sqrt{2\pi}} \frac{2^{1/4}}{2\pi} e \sqrt{\pi} = \frac{1}{2^{5/4}\pi}.$$

The differential supplies $\varepsilon^3 = n^{-3/4}$, while the remaining exponential is $e^{2\sqrt{2n}} e^{-u^2} e^\Psi$. In particular the e^{-1} in the local Bessel expression cancels the e from coefficient extraction; dropping that cancellation would change the leading normalization.

It remains to justify finite expansion and its sharp integrated remainder. Choose the Bessel expansion with $M = J + 1$. Its error is $O(|s|^{J+1}) = O(\varepsilon^{2J+2})$, and integrates to that relative order by (5.5), or by the analogous vertical Gaussian bound from (5.6). Use the resulting finite polynomial $H_J(s) = \sum_{j=0}^J h_j s^j$ in the analytic factor $H_J(s) e^{\Psi(\varepsilon, u)}$.

After cancellation in (6.2), this factor is analytic in ε near zero whenever $|\varepsilon u|$ is sufficiently small. For each fixed derivative order, the derivatives of its rational and logarithmic factors are bounded by fixed polynomials in $|u|$, uniformly for $|u| \leq C \log n$ and every real intermediate $0 \leq \varepsilon' \leq \varepsilon$. The canceled phase obeys

$$|\Psi(\varepsilon', u)| \leq C \varepsilon' (1 + |u| + |u|^3).$$

Indeed this follows from the convergent geometric expansion of $(1 + i2^{-1/4}\varepsilon' u)^{-1}$ and the logarithm expansion in s , after removing the finitely many Laurent and constant terms. Since $\varepsilon \log n \rightarrow 0$, the cubic term is at most $u^2/4$ for large n , while the other terms are bounded by $u^2/4 + O(1)$. Taylor's integral remainder through degree $2J + 1$, multiplied by e^{-u^2} , is therefore bounded by

$$C_J \varepsilon^{2J+2} (1 + |u|)^{D_J} e^{-u^2/2} \quad (6.6)$$

for a fixed D_J . Its integral is $O_J(\varepsilon^{2J+2})$, with no logarithmic loss.

The identity $s(-\varepsilon, -u) = s(\varepsilon, u)$, and the identical symmetry for Ψ , makes every odd ε coefficient an odd polynomial in u . Its integral over the symmetric segment vanishes. The Gaussian tails of each retained polynomial outside the segment are superpolynomially small. Thus the even coefficients are precisely (6.4), proving (1.1) and its remainder. The finite calculation through degree eight yields (1.2). $\square$

An independent algebraic check of the corrections

The ODE recurrence gives a useful second route to the first four coefficients, without proving existence of the expansion. Insert

$$b_n = Ce^{2\sqrt{2}n}n^{-3/4} \sum_{j \geq 0} c_j n^{-j/2}$$

into (3.3), put $z = n^{-1/2}$, and normalize each shift $n \mapsto n + d$ by the unshifted leading factor. Its contribution is

$$\exp\left(\frac{2\sqrt{2}}{z}(\sqrt{1+dz^2}-1)\right) \sum_{j \geq 0} c_j z^j (1+dz^2)^{-3/4-j/2}.$$

Expand the recurrence after multiplying by z^4 . Successive nonzero equations, with previous coefficients substituted, are

$$\begin{aligned} 8\sqrt{2}c_1 + 17/6 &= 0, \\ \sqrt{2}(9216c_2 - 6409)/576 &= 0, \\ 24\sqrt{2}c_3 + 18114133/276480 &= 0, \\ \sqrt{2}(2548039680c_4 - 13052721749)/79626240 &= 0. \end{aligned}$$

They independently recover (1.2). This is a finite symbolic identity check, not a replacement for the all-arcs and remainder proof.

7 Real inverse and discrete threshold

For a fixed $J \geq 0$ define the explicit smooth comparison

$$F_J(x) = \frac{\Gamma(x+1)^2 e^{2\sqrt{2}x}}{2^{5/4} \pi x^{3/4}} \sum_{j=0}^J c_j x^{-j/2}. \quad (7.1)$$

The correction polynomial tends to one, so F_J is positive for all sufficiently large x . Differentiated Stirling expansions give

$$(\log F_J)'(x) = 2 \log x + O_J(x^{-1/2}), \quad (7.2)$$

and therefore it is eventually strictly increasing and unbounded. Let $x_J(y)$ denote its inverse on any such eventual interval. The interval's finite initial choice has no effect for large y .

Theorem 7.1. *For $L = \log y$, let W be the positive real branch of Lambert's function and put*

$$N = \frac{L}{2W(L/(2e))}, \quad q = \log N. \quad (7.3)$$

For every fixed J , the inverse has the expansion

$$x_J(y) = N - \frac{\sqrt{2N}}{q} - \frac{1}{8} + \frac{\log 2}{8q} + \frac{1}{q^2} - \frac{1}{q^3} + O_J\left(\frac{1}{\sqrt{N}q}\right). \quad (7.4)$$

Moreover,

$$x_J(a_n) = n + O_J\left(\frac{n^{-(J+1)/2}}{\log n}\right). \quad (7.5)$$

In particular the right side of (7.4), evaluated with $y = a_n$, equals $n + O(1/(\sqrt{N} \log N))$.

Proof. The definition of W gives exactly $2N(\log N - 1) = L$. For $f(x) = 2x(\log x - 1)$ we have $f'(N) = 2q$ and $f''(N) = 2/N$. Stirling's formula in (7.1) gives

$$\log F_J(x) = f(x) + 2\sqrt{2x} + \frac{1}{4} \log x - \frac{1}{4} \log 2 + O_J(x^{-1/2}).$$

Set $\delta_0 = -\sqrt{2N}/q$. Taylor expansion at N shows that $\log F_J(N + \delta_0) - L$ equals

$$\frac{q - \log 2}{4} - \frac{2}{q} + \frac{2}{q^2} + O_J(N^{-1/2}). \quad (7.6)$$

The $\sqrt{N}$ terms cancel by the choice of δ_0 ; the quadratic term of f contributes $2/q^2$, and the linear term of the square root contributes $-2/q$. Dividing the displayed part of (7.6) by $-2q$ gives

$$\delta_1 = -\frac{1}{8} + \frac{\log 2}{8q} + \frac{1}{q^2} - \frac{1}{q^3}.$$

This is bounded. The cross term of f involving $\delta_0\delta_1$ is $O(1/(\sqrt{N}q))$; its cubic remainder is $O(1/(\sqrt{N}q^3))$. The square-root term involving δ_1 is $O(N^{-1/2})$, and its quadratic remainder is $O(1/(\sqrt{N}q^2))$. The logarithmic term changes by $O(1/(\sqrt{N}q))$. These estimates prove that

$$\log F_J(N + \delta_0 + \delta_1) - L = O_J(N^{-1/2}).$$

The derivative in (7.2) is at least q near this point for large N . For completeness, the residual and this derivative bound place a root inside a fixed sufficiently large multiple of $1/(\sqrt{N}q)$ of the point by the intermediate value theorem. Monotonicity identifies that root as $x_J(y)$, proving (7.4).

By Theorem 1.1, $\log a_n - \log F_J(n) = O_J(n^{-(J+1)/2})$. Bracketing near n with derivative at least $\log n$, or applying the mean value theorem after that bracketing, proves (7.5). The final statement follows by combining this estimate with (7.4); $N \sim n$. $\square$

Constructing further fixed inverse orders

All higher fixed inverse accuracy is constructive. One may use the known finite expansion of $\log F_J$ and Newton reversion, with the error divided by a derivative bounded below by $\log x$. Here is an explicit procedure with a quantitative stopping criterion. For any fixed $B > 0$, Stirling's series and the finite logarithmic expansion of the correction polynomial give an explicit elementary function $T(x)$ with

$$T(x) = \log F_J(x) + O_{J,B}(x^{-B}), \quad T'(x) = 2 \log x + O_J(x^{-1/2}), \quad T''(x) = O_J(1/x).$$

Retain sufficiently many half-powers (and Stirling terms) to ensure that remainder. The derivative statements follow by differentiating the finite expression, rather than differentiating an unspecified O term. The corresponding remainder for the original smooth function follows from the usual Stirling remainder and the convergent logarithm of the finite correction polynomial near one.

Starting at $X_0 = N$, iterate the entirely explicit rule

$$X_{k+1} = X_k - \frac{T(X_k) - L}{T'(X_k)}. \quad (7.7)$$

The root of $T(x) = L$ is within $O(\sqrt{N}/\log N)$ of N and within $O_{J,B}(N^{-B}/\log N)$ of $x_J(y)$. On that neighborhood the Newton error satisfies

$$|e_{k+1}| \leq \frac{C_J}{N \log N} |e_k|^2.$$

This follows from Taylor's theorem, the bound on T'' , and the lower bound on T' . For every fixed desired power of N^{-1} , a finite number of iterations suffices. Choosing B larger than that desired power yields a finite expression to that accuracy. Expanding these finite expressions gives the customary formal inverse series. For an inverse valid at sequence values, increasing J in (7.5) supplies any requested fixed half-power accuracy divided by $\log n$. This procedure does not determine unspecified effective error constants or threshold locations.

Theorem 7.2. *Define the actual integer crossing by*

$$I(y) = \min\{n \geq 0 : a_n \geq y\}.$$

For every fixed J , there are $K_J > 0$ and Y_J such that for $y \geq Y_J$, with $x = x_J(y)$,

$$\left\lceil x - K_J \frac{x^{-(J+1)/2}}{\log x} \right\rceil \leq I(y) \leq \left\lceil x + K_J \frac{x^{-(J+1)/2}}{\log x} \right\rceil. \quad (7.8)$$

Eventually the two displayed integers are equal or adjacent.

Proof. Appending a zero row and column injects the $n \times n$ class into the $(n+1) \times (n+1)$ class. The matrix with a single one in the new bottom-right position is outside the image, so a_n is strictly increasing. It is unbounded, for example because it contains every permutation matrix. Thus $I(y)$ exists.

Put $\alpha = (J+1)/2$, $e(t) = t^{-\alpha}/\log t$, and $z_n = x_J(a_n)$ for sufficiently large n . Equation (7.5) says $|z_n - n| \leq Ce(n)$. For large y all relevant indices are in this range, and strict monotonicity gives

$$I(y) = \min\{n : z_n \geq x\}.$$

Since $e(n) \rightarrow 0$, neighboring integer bounds first give $I(y) = x + O(1)$. Explicitly, for all sufficiently large indices $|z_n - n| < 1$; hence indices below $x - 1$ cannot cross, whereas any integer at least $x + 1$ does. Uniformly for integers $m = x + O(1)$, $e(m)/e(x) \rightarrow 1$, so $e(m) \leq 2e(x)$ eventually.

Let $k = I(y)$. Then $x \leq z_k \leq k + Ce(k)$, so $k \geq x - 2Ce(x)$. Since k is an integer, $k \geq \lceil x - K_J e(x) \rceil$ whenever $K_J \geq 2C$. For the upper bound, choose $K_J > 2C$ and put $m = \lceil x + K_J e(x) \rceil$. Here $m = x + O(1)$, and

$$z_m \geq m - Ce(m) \geq x + (K_J - 2C)e(x) \geq x.$$

Thus $k \leq m$. This proves both sides, including at exact thresholds $y = a_n$ and arbitrarily close to them. Finally $2K_J e(x) < 1$ eventually, so the two ceilings differ by at most one. $\square$

The enclosure is an asymptotic theorem with existential constants. It is not a finite-input certified rounding algorithm. In general one cannot round an asymptotic real inverse in a universally correct way: y can be arbitrarily close to a discrete threshold. Formula (7.4) alone therefore gives $I(y)$ equal to its displayed real expression plus $O(1)$, while (7.8) supplies the stronger rounding-safe statement when the comparison and error allowance are retained.

Remark 7.3 ([write] The inverses against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`; below its h , f , E and t of `p0:thm:core-reversion` are written h_{vol} , f_{vol} , ξ and τ , to keep them apart from the source's $h(t)$, $f(x)$, $E(s)$ and $t = xy$.

- (a) N is an instance of `p0:prop:factorial-core`, verbatim, with $(\kappa, d, L) := (2, -2, L)$: the phase is $X(2 \log X - 2) = 2X(\log X - 1)$, the solution is $v = W_0((L/2)e^{-1}) = W_0(L/(2e))$ and $X = e^{v+1} = L/(2v) = N$ (because $ve^v = L/(2e)$), and the side condition $2 \log X - 2 > 0$ is $v > 0$, that is $L > 0$, that is $y > 1$. Hence $q = \log N = v + 1$, and the core slope $\kappa(v + 1) = 2q$ is the $f'(N) = 2q$ of the proof of Theorem 7.1. This is the core of `p0:rem:factorial-core` (the double factorial normalization $\frac{s}{2}(\log s - 1)$) at the target $L/4$. The source's positive real branch is W_0 .
- (b) The expansion (7.4) truncates an instance of the formal theorem `p0:thm:core-reversion`, after the change of variables $x = N(1 + \xi)$, $\tau = N^{-1/2}$ and division by $2N$. By $L = 2N(q - 1)$ and Stirling's series, $\log F_J(x) = L$ becomes

$$q\xi + h_{\text{vol}}(\xi) + f_{\text{vol}}(\tau, \xi) = 0, \quad h_{\text{vol}}(\xi) = (1 + \xi) \log(1 + \xi) - \xi,$$

$$f_{\text{vol}}(\tau, \xi) = \sqrt{2}\tau(1 + \xi)^{1/2} + \frac{\tau^2}{8}(q + \log(1 + \xi) - \log 2) + \tau^3\rho(\tau, \xi),$$

where $\tau^3\rho$ collects $\frac{\tau^2}{2} \log \sum_{j \leq J} c_j x^{-j/2}$ (a power series without constant term in $x^{-1/2} = \tau(1 + \xi)^{-1/2}$) and the terms $\frac{\tau^2}{2} \cdot \frac{1}{6x} + \dots$ of Stirling's series. This is the master equation of the theorem over $R = \mathbb{Q}(\sqrt{2})[\log 2, q, q^{-1}]$, with $\Lambda = q$, $h_{\text{vol}} \in u^2 R[[u]]$ (the h of `p0:rem:core-instances`) and $f_{\text{vol}} \in \tau R[[\tau, u]]$. Its triangular recurrence gives $\xi = e_1\tau + e_2\tau^2 + O(\tau^3)$ with $qe_1 + \sqrt{2} = 0$ and $qe_2 + \frac{1}{2}e_1^2 + \frac{\sqrt{2}}{2}e_1 + \frac{q}{8} - \frac{\log 2}{8} = 0$, that is $e_1 = -\sqrt{2}/q$ and $e_2 = -\frac{1}{8} + \frac{\log 2}{8q} + \frac{1}{q^2} - \frac{1}{q^3}$; so $N\xi = e_1\sqrt{N} + e_2 + O(N^{-1/2})$ has exactly the δ_0 and δ_1 of the proof. Two qualifications. The half power $(1 + \xi)^{1/2}$ and the logarithm put f_{vol} outside the Bell form (`p0:eq:core-bell`), which needs integer exponents $-\mu_k$, so only the general part of the theorem applies; and because f_{vol} contains $\tau^2 q/8$, the coefficient e_2 has the constant term $-\frac{1}{8}$: the e_n are still polynomials in q^{-1} , as in `p0:rem:core-instances`, but not all without constant term. (The shift $\xi = \xi' - \tau^2/8$ removes q from the perturbation, so $e_n \in q^{-1}R_0[q^{-1}]$ for $n \neq 2$ and $e_2 + \frac{1}{8} \in q^{-1}R_0[q^{-1}]$, with $R_0 = \mathbb{Q}(\sqrt{2})[\log 2]$.) (Corrected after the independent check of 6 October 2026. The first wording read “the coefficients e_n are Laurent polynomials in q , not polynomials in q^{-1} as in `p0:rem:core-instances`”; that contrast is false: $e_1, \dots, e_4$, computed with the full Stirling series, contain no positive power of q , and only the constant term of e_2 differs.) The theorem is formal: the remainder $O_J(1/(\sqrt{N}q))$ in (7.4), (7.5) and the Newton procedure (7.7) are the source's own analysis, not supplied by the volume.

- (c) Theorem 7.2 is not an instance of `p0:thm:staircase` as proved (its proof uses no interpolation), but it follows from part (1) of that theorem once an admissible interpolation is supplied, and here is one. The hypotheses of `p0:def:three-inverses` hold for $A_n = a_n$: a_n is strictly increasing for $n \geq 0$ and unbounded (proof of Theorem 7.2), and $I(y) = N_*(y)$ for every

n_1 once $y > a_{n_1-1}$. Fix J ; let C , $e(\cdot)$ and $z_n = x_J(a_n)$ be as in that proof, and choose n_1 so large that for $n \geq n_1 - 1$ the value a_n lies in the domain of x_J , $|z_n - n| \leq Ce(n)$, and e is decreasing. Let φ be the piecewise-linear function on $[n_1, \infty)$ with $\varphi(n) = z_n$, and $\mathcal{A} := F_J \circ \varphi$. [write] *Claim: $\mathcal{A}$ is admissible, and $|\nu(y) - x_J(y)| \leq 2Ce(x_J(y))$ for all large y , where $\nu = \mathcal{A}^{-1}$.* Indeed z_n is strictly increasing (x_J and a_n are), so φ is continuous, strictly increasing and unbounded with values in the interval on which F_J increases; $\mathcal{A}$ is therefore continuous and strictly increasing, and $\mathcal{A}(n) = F_J(x_J(a_n)) = a_n$. For $y \geq a_{n_1}$ put $x = x_J(y)$ and $w = \nu(y)$, so $\varphi(w) = x$. If $n \leq w \leq n+1$, then $\varphi(w) - w$ is a convex combination of $z_n - n$ and $z_{n+1} - (n+1)$, so $|x - w| \leq C \max(e(n), e(n+1)) = Ce(n)$. As $y \rightarrow \infty$, $w \rightarrow \infty$ and $|x - w| \rightarrow 0$, so $e(n)/e(x) \rightarrow 1$ and $|x - w| \leq 2Ce(x)$ eventually. By `p0:thm:staircase(1)`, $I(y) = \lceil \nu(y) \rceil$, and since the ceiling is nondecreasing, $\lceil x - 2Ce(x) \rceil \leq I(y) \leq \lceil x + 2Ce(x) \rceil$: this is (7.8) with $K_J = 2C$. Part (2) of the theorem adds that $I(y)$ equals the common value whenever the two ceilings agree.

- (d) *Rounding at the sequence values is an instance of `p0:thm:staircase(3)`: with $\mathcal{A}$ as in (c), $\nu(a_n) = n$, and the right side x_* of (7.4) at $y = a_n$ satisfies $|x_* - n| = O(1/(\sqrt{N} \log N)) < \frac{1}{2}$ for large n (Theorem 7.1), so $n = \lfloor x_* + \frac{1}{2} \rfloor$ beyond an existential onset. (The write's evaluations of $x_* - n$ at $n = 20, 100, 1000, 4000$ are $-0.0103, -0.0036, -0.00088, -0.00038$; end of Section 8.)*

No novelty is claimed for these inversions.

8 Exact fixtures and reproducibility

The seventeen values in the inspected OEIS record, beginning at $n = 0$, are reproduced below as finite numeric fixtures. The companion checks every value against the exact recurrence.

n	a_n
0	1
1	2
2	16
3	265
4	7343
5	304186
6	17525812
7	1336221251
8	129980132305
9	15686404067098
10	2297230134084416
11	400977650310256537
12	82188611938415464231
13	19536244019455339261970
14	5328019975275896220786388
15	1651867356348327784988233291
16	577522171260292028520919811777

The following independent counting construction does not use the ODE. For a fixed number n of columns, let a state (k, ℓ) record the numbers of columns currently of degree zero and one. A new

row has zero, one, or two ones. Its transitions, with multiplicities, are

new state	multiplicity
(k, ℓ)	1
$(k-1, \ell+1)$	k
$(k, \ell-1)$	ℓ
$(k-2, \ell+2)$	$\binom{k}{2}$
$(k-1, \ell)$	$k\ell$
$(k, \ell-2)$	$\binom{\ell}{2}$

Start at $(n, 0)$, repeat for n rows, and sum the state weights. Invalid states are omitted. A more literal check retains the entire labeled column-degree tuple. Small binary matrices can also be exhaustively checked by their n^2 -bit masks. These constructions supply checks with genuinely different state representations.

The source archive contains the LaTeX source, deterministic PDF and ZIP builders, a manifest, finite numeric fixtures, and bounded companion programs. The core exact arithmetic uses standard-library Python integers and rational numbers; the finite symbolic coefficient calculations also use only the standard library. The companion’s usage guide specifies implemented input bounds, commands, and precise test coverage. The code uses explicit exceptions and test assertions that remain active with Python optimization. No network access is needed to reproduce the packaged computations.

Finite exact agreement verifies implementations and identities, not the asymptotic theorem. Conversely, the proof gives no effective finite- n error constant. Any decimal diagnostic in the companion is uncertified floating-point evidence and must not be read as an interval bound or proof. The release README records build requirements and replay commands; the verification receipt reports the actual completed checks.

[write] The package here, the intake reruns and the write’s computations, 6 October 2026.

“The source archive”, “the companion”, “the release README” and “the verification receipt” above are the delivered ones. In this report the companion programs are `code/companion-exact_matrix.py` and `code/companion-symbolic_coefficients.py`, with their tests `code/companion-test_exact_matrix.py` and `code/companion-test_symbolic_coefficients.py` (delivered `companion/...`); the usage guide is `companion-USAGE.md`; the release tools `build_pdf.py`, `make_zip.py`, `release_tools.py` and `test_release.py` are in `code/`; `SOURCE_PROVENANCE.json` and `verification_receipt.json` are in `data/` with the frozen outputs; the manifest and the PDF are not shipped, and the receipt’s hashes refer to the delivered **Report168.tex** and PDF. The programs read their data from `data/` beside the directory that holds them and the tests import the programs by their delivered names, so they run only on a copy with the delivered names (the report README gives two routes); the release tools need the whole delivered package and POSIX. *Reruns on copies at the write* (Windows, Python 3.14.4): the `verify` output and the order-4 symbolic output equal `data/exact_verification.json` and `data/symbolic_order4.json` once Windows line endings are converted; the 35 companion tests pass, both from a fresh extraction and from the shipped files restored to their delivered names; 10 of the 11 release tests error (no `os.mkfifo`; “POSIX O_NOFOLLOW is required”). The symbolic program at order 6 gives $c_5 = -\frac{3563067617707\sqrt{2}}{342456532992} = -14.7140\dots$ and $c_6 = \frac{109526487454949291}{2465687037542400} = 44.4202\dots$

The write’s own computations (programs of the write, not shipped). (i) a_n for $n \leq 4$ by enumerating all 2^{n^2} matrices, for $n \leq 7$ by a row-by-row count over the full vector of column degrees, and for $n \leq 16$ as $(n!)^2[x^n y^n] e^{x+y-xy} F_{154}$ from the A062154 formula (Remark 2.1): all equal the OEIS terms. The same series gives the 17 listed terms of A062156 and rows 1–3 of A062154, and $(1-t)^{-1/2}e^{-t/2}$ gives A001499 for $n \leq 16$; exact rationals throughout. (ii) The Bessel diagonal (2.3), expanded exactly to t^{80} , satisfies (3.3) for $n \leq 77$; the recurrence was then run in 120-digit arithmetic to $n = 4000$. With R_n the ratio of a_n to the leading term of (1.1), $n^{5/2}(R_n - \sum_{j \leq 4} c_j n^{-j/2})$ is $-11.33, -13.44, -14.05$ at $n = 100, 1000, 4000$ (compare $c_5 = -14.71$), and after subtracting also the c_5 and c_6 terms the remainder times $n^{7/2}$ is $-116.4, -127.4$,

−133.7 at $n = 250, 1000, 4000$. (iii) $\log a_n$ minus the expansion (1.3), times $n^{3/2}$: −1.41, −1.63, −1.69 at $n = 100, 1000, 4000$. (iv) The right side of (7.4) at $y = a_n$, minus n , times $\sqrt{N}q$: from −0.151 to −0.199 at $n = 20, 50, 100, 1000, 4000$; and $x_4(a_n) - n$, computed as the root of $\log F_4(x) = \log a_n$, times $n^{5/2} \log n$: from −4.3 at $n = 20$ to −7.0 at $n = 4000$, consistent with the heuristic limit $c_5/2$. These are floating-point observations at finitely many n , consistent with the orders proved; they certify nothing.

[write] Independent check of the write, 6 October 2026. An adversarial check made by the intake after the write (commit 3834812cd) re-derived the statements marked [write] with its own code. *Remark 2.1.* The identity $F = e^{x+y-xy} F_{154}$ and its combinatorial reading were re-derived; $(n!)^2 [x^n y^n] e^{x+y-xy} F_{154}$ equals all 17 terms of A197458, the same series all 46 data terms of A062154, the Bessel diagonal all 17 terms of A062156, and $(1-t)^{-1/2} e^{-t/2}$ and the Bricard recurrence the terms of A001499; brute force over $\{0, 1, 2\}$ -matrices gives rows 1–3 of A062154 and $T(4, 4) = 4842$, and over binary matrices 1, 2, 16, 265, 7343, 304186 for $n \leq 5$ and every $A_{m,n}$ with $m, n \leq 5$. The proof of $A001499(n)/(n!)^2 \sim e^{-1/2}/\sqrt{\pi n}$ was checked step by step, and the names and revisions match the live entries. *Remark 7.3.* Items (a), (c) and (d) were re-derived, and the numbers of (d) reproduced from (3.3) at 60 digits. In (b), SymPy confirms the master equation and $e_1 = -\sqrt{2}/q$, $e_2 = \delta_1$, but also that e_3 and e_4 have q -exponents −1 to −5 and −1 to −7; the sentence on the shape of the e_n is corrected above. (When the check was applied, the reversion was recomputed with the remainder $\tau^3 \rho$ replaced by a generic q -free double series: $e_1, \dots, e_5$ have no positive power of q , and only e_2 has a constant term, $-\frac{1}{8}$.) *Lemma 9.1.* Sections 5 and 6 were re-read: the diagonal enters only through the circle bound, the local expansion on arcs inside a fixed sector, the size of $r^{-n} G(r)$ and the analyticity needed for the deformation. *Proposition 9.2.* An independent implementation of (6.4) reproduces $c_1, \dots, c_4$ to 25 digits and gives $c_1^{(d)} - c_1 = -\frac{\sqrt{2}}{2} d(d-1)$ for $d = 1, 2, 3$. Counts to $n = 800$ from a column-degree recursion (checked against exact counts): the ratio $A_{n+2,n}/((n+2)! n! b_n)$ is 0.887193, 0.915356, 0.937392, 0.954218 at $n = 100, 200, 400, 800$, within $O(n^{-5/2})$ of the five-term prediction; the proportion without a zero line is 0.17547, 0.16336, 0.15496, 0.14912, and its difference from $e^{-2}(1 + 2\sqrt{2}/\sqrt{n})$ (the two-term prediction) times n is a stable 0.087 to 0.100; $E[Z_r] = E[Z_c]$ and $E[Z_r Z_c]$ rise from 0.877 and 0.874 to 0.953 and 0.952, toward 1. The marked generating functions equal brute-force joint laws for $n \leq 5$, and $n^{5/2}(R_n - \sum_{j \leq 4} c_j n^{-j/2}) = -11.331$ at $n = 100$ independently of the differential equation. No other mathematical error was found. One sentence was corrected, with a dated note keeping the first wording: Remark 7.3(b) said that the e_n are “Laurent polynomials in q , not polynomials in q^{-1} ”; they are polynomials in q^{-1} , and the real difference from `p0:rem:core-instances` is the constant term $-\frac{1}{8}$ of e_2 .

9 Questions for further research

The exact bivariate function (2.1) suggests several concrete extensions that are not proved here.

[write] Added 6 October 2026, batch 108. Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated as an open question at the end of this section, with its source, the argument offered and what is missing. The source asserts nothing beyond its proofs; its non-claims are collected in Section 1.1. The intake found no false claim in the source and the write found none either. The write proves two cases of the first two directions below (Lemma 9.1, Proposition 9.2).

Rectangular aspect ratios. For $m \neq n$, the coefficient $m!n! [x^m y^n] F(x, y)$ invites a bivariate saddle analysis, or a uniform analysis of Bessel functions whose order grows with the dimensions. A useful first target is a uniform leading formula when m/n stays in a compact subinterval of $(0, \infty)$, followed by the transition as the aspect ratio approaches one on a shrinking scale. The balanced Bessel diagonal alone does not supply that uniformity.

Marked components and distributions. Separate variables can mark isolated vertices, balanced and unbalanced paths, or cycles in the connected-component exponent. Differentiating an appropriate

marked coefficient expansion could yield means, covariances, and limit laws for a uniformly chosen matrix. Such claims require complex-uniform marked estimates and control of the normalization; the unmarked expansion does not by itself prove them.

Effective errors and onset. Explicit constants in the Bessel remainder, arc estimates, and Taylor majorant could turn the eventual two-ceiling theorem into a certified finite-input enclosure. This would require a quantitative onset and verified error allowances, rather than fitting the constants from finite exact values.

Exponentially small terms. The present minor-arc bound is sufficient for every fixed algebraic order, but does not classify secondary saddles or Stokes effects of the Bessel factor. Determining late coefficient growth, optimal truncation, and exponentially small contributions is a separate problem. No complete transseries is supplied by the current proof.

Two cases proved at the write

The proof of Theorem 1.1 uses the diagonal G only through two properties, and other diagonals of the same family have them.

Lemma 9.1 ([write] Transfer of the proof of Theorem 1.1). *Let $\tilde{G}(t) = \sum_{n \geq 0} \tilde{g}_n t^n$ have nonnegative coefficients and radius of convergence at least 1, and suppose:*

- (A) *there is an absolute C_0 with $|\tilde{G}(re^{i\theta})| \leq C_0 \tilde{G}(r) e^{-D(r,\theta)}$ for $0 < r < 1$ and $-\pi \leq \theta \leq \pi$, with D as in (5.1);*
- (B) *there are a sector $|\arg s| \leq \sigma$, $\sigma > 0$, and a formal series $\tilde{H}(s) = \sum_{j \geq 0} \tilde{h}_j s^j$ such that for every $M \geq 1$, uniformly as $s \rightarrow 0$ in the sector,*

$$\tilde{G}(1-s) = \frac{e^{-1}}{\sqrt{2\pi}} e^{2/s} \left(\sum_{j=0}^{M-1} \tilde{h}_j s^j + O_M(|s|^M) \right).$$

Then for every $J \geq 0$

$$\tilde{g}_n = \frac{e^{2\sqrt{2n}}}{2^{5/4} \pi n^{3/4}} \left(\sum_{j=0}^J \tilde{c}_j n^{-j/2} + O_J(n^{-(J+1)/2}) \right), \quad \tilde{c}_j = \mathcal{M}\left([\varepsilon^{2j}] \tilde{H}(s) e^{\Psi(\varepsilon, u)}\right),$$

and in particular $\tilde{c}_0 = \tilde{h}_0$.

Proof. Sections 5 and 6 use G only through the circle bound (5.2), the local expansion (4.4) in a sector about the positive axis (every point used has $|\arg s| = O(n^{-1/4} \log n)$), and the consequence $r^{-n} G(r) = O(e^{2\sqrt{2n}})$ of (4.4) on the positive axis; and the coefficient rule uses only the coefficients of H , which depends on s alone, as the parity argument requires. Hypotheses (A) and (B) are these properties for $\tilde{G}$ and $\tilde{H}$ (with the constant e of (5.2) replaced by C_0 ; (B) on the positive axis gives $\tilde{G}(r) = O(e^{2/s_0})$ and so $r^{-n} \tilde{G}(r) = O(e^{2\sqrt{2n}})$). Hence the argument from (5.4) to the end of the proof of Theorem 1.1 applies word for word with G , H , b_n , c_j replaced by $\tilde{G}$, $\tilde{H}$, $\tilde{g}_n$, $\tilde{c}_j$. Since Ψ has no ε^0 term, $\tilde{c}_0 = \tilde{h}_0$. $\square$

Proposition 9.2 ([write] Fixed offsets and empty lines). (a) For each fixed integer $d \geq 0$ and every $J \geq 0$, $A_{n+d,n} = A_{n,n+d}$ and

$$\frac{A_{n+d,n}}{(n+d)!n!} = \frac{e^{2\sqrt{2}n}}{2^{5/4}\pi n^{3/4}} \left(\sum_{j=0}^J c_j^{(d)} n^{-j/2} + O_{d,J}(n^{-(J+1)/2}) \right),$$

where $c_j^{(d)}$ is given by (6.4) with H replaced by

$$H_d(s) = (1-s)^{-d/2} p(s)^{-1/2} \exp\left(\frac{p(s)-1}{s} - \frac{1}{2} + \frac{s}{2}\right) \sum_{k \geq 0} \gamma_k^{(d)} \left(\frac{s}{p(s)}\right)^k, \quad \gamma_k^{(d)} = (2k-1)!! \beta_k^{(d)},$$

$\sum_k \beta_k^{(d)} w^{2k} = T_d(1-w^2/2)(1-w^2/4)^{-1/2}$ and T_d the Chebyshev polynomial $((-1)!! = 1)$. In particular $c_0^{(d)} = 1$, $c_j^{(0)} = c_j$, and $c_1^{(d)} = c_1 - \frac{\sqrt{2}}{2}d(d-1) = -\frac{\sqrt{2}}{96}(17+48d(d-1))$.

(b) Let Z_r and Z_c be the numbers of zero rows and of zero columns of a matrix drawn uniformly from the a_n . As $n \rightarrow \infty$, (Z_r, Z_c) converges in distribution to a pair of independent Poisson variables of mean 1, with convergence of all mixed moments. In particular the proportion of the a_n matrices with no zero row and no zero column tends to e^{-2} .

Proof. (a) Transposition gives $A_{n+d,n} = A_{n,n+d}$. In (2.1), the monomials $x^{k+d}y^k$ of $e^{xh(t)}e^{yh(t)}$ carry $h^{2k+d}/((k+d)!k!)$, so, as for (2.3),

$$G_d(t) := \sum_{n \geq 0} \frac{A_{n+d,n}}{(n+d)!n!} t^n = P(t)S_d(t), \quad S_d(t) = \sum_{k \geq 0} \frac{t^k h(t)^{2k+d}}{k!(k+d)!} = t^{-d/2} I_d(q(t)),$$

with $I_d(z) = \sum_k (z/2)^{2k+d}/(k!(k+d)!)$. (A). On $|t| = r$, $|h(t)| = |\frac{1}{2} + \frac{1}{2(1-t)}| \leq h(r)$, so $|S_d(t)| \leq S_d(r)$; with the bound $|P(t)| \leq e P(r) e^{-D(r,\theta)}$ from the derivation of (5.2), (A) holds with $C_0 = e$. (B). For integer d , $I_d(z) = \frac{1}{\pi} \int_0^\pi e^{z \cos v} \cos(dv) dv$, and $\cos(dv) = T_d(\cos v) = T_d(1-w^2/2)$ for $w = 2 \sin(v/2)$. The proof of Lemma 4.1 applies with $(1-w^2/4)^{-1/2}$ replaced by $T_d(1-w^2/2)(1-w^2/4)^{-1/2}$: on $[0, 1]$ expand it as $\sum_{k < M} \beta_k^{(d)} w^{2k} + O(w^{2M})$; on $[1, 2]$ it is still integrable ($|T_d| \leq 1$ there); and $\frac{1}{\pi} 2^{k-1/2} \Gamma(k + \frac{1}{2}) = \frac{(2k-1)!!}{\sqrt{2\pi}}$. Hence $I_d(q) = \frac{e^q}{\sqrt{2\pi q}} (\sum_{k < M} \gamma_k^{(d)} q^{-k} + O(|q|^{-M}))$ uniformly in $|\arg q| \leq \pi/2 - \delta$, and the substitution of (4.3)–(4.5), with the extra factor $t^{-d/2} = (1-s)^{-d/2}$, gives (B) with $\tilde{H} = H_d$. For $d = 0$, $T_0 = 1$ and $\gamma_k^{(0)} = \gamma_k$, so $H_0 = H$. Lemma 9.1 gives the expansion and $c_0^{(d)} = H_d(0) = 1$. For $c_1^{(d)}$: $T_d(1-x) = 1 - d^2 x + O(x^2)$ gives $\beta_1^{(d)} = \frac{1}{8} - \frac{d^2}{2}$, so $\gamma_1^{(d)} - \gamma_1 = -\frac{d^2}{2}$, and $(1-s)^{-d/2} = 1 + \frac{d}{2}s + O(s^2)$; hence $[s^1]H_d - [s^1]H = -\frac{d(d-1)}{2}$. In (6.4) at $j = 1$ the amplitude enters only through $h_0 E_2 + \sqrt{2} h_1$ (as $s = \sqrt{2}\varepsilon^2 + O(\varepsilon^3)$ and E_1 is odd in u), and $h_0 = 1$ for both, so $c_1^{(d)} - c_1 = -\frac{\sqrt{2}}{2}d(d-1)$; with $c_1 = -\frac{17\sqrt{2}}{96}$ this is the stated value.

(b) A zero row or column is an isolated vertex, the term $x + y$ of the component table. Marking zero rows by $\omega_1 \geq 0$ and zero columns by $\omega_2 \geq 0$ replaces the factor $e^{(x+y)h(t)}$ of (2.1) by $e^{x(h(t)-1+\omega_1)+y(h(t)-1+\omega_2)}$, so

$$\tilde{G}_\omega(t) := P(t) \sum_{k \geq 0} \frac{t^k (h(t) - 1 + \omega_1)^k (h(t) - 1 + \omega_2)^k}{(k!)^2} = \sum_{n \geq 0} b_n \mathbb{E}_n[\omega_1^{Z_r} \omega_2^{Z_c}] t^n.$$

(A). $h(t) - 1 = t/(2(1-t))$, so on $|t| = r$, $|h(t) - 1 + \omega_i| \leq h(r) - 1 + \omega_i$; as in (a), (A) holds with $C_0 = e$. (B). The sum is $I_0(\tilde{q})$ with $\tilde{q}^2 = 4t(h-1+\omega_1)(h-1+\omega_2)$ (I_0 is even, so the choice of root

is immaterial). With $t = 1 - s$ and $h - 1 = (1 - s)/(2s)$,

$$\tilde{q} = \frac{(1 - s)^{3/2}}{s} \left(1 + \frac{2s\omega_1}{1 - s}\right)^{1/2} \left(1 + \frac{2s\omega_2}{1 - s}\right)^{1/2} = \frac{p(s)}{s} + \rho(s),$$

with ρ analytic at 0 and $\rho(0) = \omega_1 + \omega_2 - 2$, because $(1 - s)^{3/2} - p(s) = -2s\sqrt{1 - s}$ and the product of the square roots is $1 + s(\omega_1 + \omega_2) + O(s^2)$. In a narrow sector $|\arg \tilde{q}| \leq \pi/2 - \delta$, so Lemma 4.1 and the computation of (4.4) give (B) with

$$\tilde{H}_\omega(s) = (p + s\rho)^{-1/2} \exp\left(\frac{p-1}{s} - \frac{1}{2} + \frac{s}{2} + \rho(s)\right) \sum_{k \geq 0} \gamma_k \left(\frac{s}{p + s\rho}\right)^k, \quad \tilde{H}_\omega(0) = e^{\omega_1 + \omega_2 - 2}.$$

Lemma 9.1 with $J = 0$ and Theorem 1.1 give $\mathbb{E}_n[\omega_1^{Z_r} \omega_2^{Z_c}] \rightarrow e^{(\omega_1 - 1) + (\omega_2 - 1)}$ for all $\omega_1, \omega_2 \geq 0$: the joint generating function of two independent Poisson(1) variables. Convergence on $[0, 1]^2$ gives convergence in distribution (continuity theorem for probability generating functions). Convergence at $(2, 2)$ bounds $\mathbb{E}_n[2^{Z_r + Z_c}]$ for large n ; since $z^i \leq C_i(3/2)^z$, every mixed moment $Z_r^i Z_c^j$ is uniformly integrable, and the moments converge. $\square$

The write's exact computations agree: brute force confirms the generating functions of (a) for $d = 1, 2$, $n \leq 3$, and of (b) at $\omega = (0, 0)$ for $n \leq 4$; the ratio $A_{n+2,n}/((n+2)!n!b_n)$ is 0.8395, 0.8767, 0.9070 at $n = 40, 80, 160$, with $\sqrt{n}$ times its distance from 1 equal to $-1.01, -1.10, -1.18$ (the proposition gives the limit $c_1^{(2)} - c_1 = -\sqrt{2}$); for $d = 1$ the ratio is 1.0067, 1.0040, 1.0023 ($c_1^{(1)} = c_1$); and the proportion with no zero line is 0.2290, 0.2002, 0.1804, 0.1718 at $n = 20, 40, 80, 120$, approaching $e^{-2} = 0.1353$ slowly.

The open questions

1. **Rectangular aspect ratios** (source: the paragraph so named above). Proved: Theorem 1.1 for $m = n$, and (write) Proposition 9.2(a) for $|m - n| = d$ fixed, where the offset diagonal is a Bessel function of fixed order d . Open: a uniform formula for $A_{m,n}$ with m/n in a compact subinterval of $(0, \infty)$, and the transition as $m/n \rightarrow 1$ with $|m - n| \rightarrow \infty$. The source suggests a bivariate saddle analysis or Bessel functions of growing order; missing are the uniform estimates.
2. **Marked components and distributions** (source: the paragraph so named above). Proved (write): Proposition 9.2(b), an independent Poisson(1) limit for zero rows and zero columns, whose marks enter only the Bessel argument. Open: means, variances and limit laws for the numbers of balanced paths, unbalanced paths and cycles. A mark on balanced or on unbalanced paths changes the coefficient of the pole in $t/(1 - t)$ or in the Bessel argument and so moves the saddle point, and a mark on cycles changes the power $(1 - t)^{-1/2}$ of P ; either way hypothesis (B) of Lemma 9.1 fails as stated, and, as the source says, complex-uniform marked estimates and control of the normalization are needed.
3. **Effective errors and onset** (source: the paragraph so named above). Open: explicit constants in Lemma 4.1, the arc and connector estimates and the Taylor majorant (6.6), which would turn Theorem 7.2 into a certified finite-input enclosure and make Remark 7.3(d) effective. Fitting constants to finite data does not do this.

4. **Exponentially small terms** (source: the paragraph so named above). Open: the secondary saddles and Stokes effects of the Bessel factor, the growth of c_j (the values $c_4 = 5.12\dots$, $c_5 = -14.71\dots$, $c_6 = 44.42\dots$ grow in size, which says nothing about the late terms), optimal truncation and a transseries.
5. **The literature boundary** (source: Section 10). Open: the exact scope of Goulden and Jackson's Problem 3.4.15, Knuth's solution and a specialization of Greenhill, McKay and Wang to the sum over degree vectors; by Remark 2.1 the generating function itself follows from a formula of OEIS A062154. The source's bounded search did not find the scalar expansion in print; the write made no further search.
6. **At most k ones**, $k \geq 3$ (added by the write, not a claim of the source). A283500 lists A197458 as the column $k = 2$ of the square family with at most k ones per line; for $k \geq 3$ the components are no longer paths and cycles, and no analogue of (2.1) or of Theorem 1.1 is known to the write. The regime $k = n/2$ is treated by Ordentlich, Parvaresh and Roth [9].

10 Literature coverage and claim boundaries

The inspected A197458 record defines the same square binary class and provides a multivariate recurrence and Java implementation [1]. Its revision is number 25, dated 8 April 2020; the source retrieval on 3 October 2026 found no asymptotic formula in that record. Mathar's seven-page note gives recurrences, numerical tables, and a Maple implementation; its Table 2 contains the relevant rectangular family [2]. The linked original discussion gives the recurrence rather than an asymptotic [3].

Closely related component EGFs are older. OEIS A062154 records a bivariate EGF for matrices over $\{0, 1, 2\}$ with line sums one or two, and it and the square companion A062156 cite Goulden and Jackson, *Combinatorial Enumeration*, Problem 3.4.15 [4]. The printed exercise and its answer were not directly inspected. Their exact scope remains an explicit source gap; the indirect citation must not be converted into a claim about an unread theorem. The exactly-two binary family A001499 is also classical, and the retrieved OEIS entry records a leading equivalent and older references [5]. This nearby class is not A197458, but it precludes any suggestion that degree-two bipartite enumeration itself is new.

Greenhill, McKay, and Wang's uniform asymptotics for prescribed sparse row and column degree sequences include bounded degrees at most two [6]. Their theorem does not explicitly perform the sum over all vectors defining a_n , but it provides classical machinery from which a leading specialization can be approached. Caizergues and de Panafieu give a general exact bounded-degree bipartite generating function, allowing different degree weights on the two shores [7]; this includes the finite degree set of the present class. The retrieved paper does not state the scalar asymptotic proved here. These facts warrant crediting general enumeration methods rather than claiming a new enumeration principle.

The present work supplies an elementary explicit specialization, a sharp all-fixed-orders calculation, four displayed corrections, and an inverse with a safe discrete interpretation. A bounded search of retrieved sources did not locate the particular scalar expansion; that is not a historical-priority certificate. Older degree-two terminology and the unread textbook exercise leave uncertainty. No claim of first discovery is needed for any mathematical statement in this report.

The primary analytic scope is also deliberately narrow: all orders are fixed before taking the

limit; no complete beyond-all-orders transseries, convergent coefficient expansion, effective global remainder constant, or universally correct rounded inverse is asserted. The theorem and exact checks remain valid independently of whether the scalar equivalent appears elsewhere in the literature.

[write] **The OEIS records and the literature at the write, 6 October 2026.** A197458 is still at revision #25, without an asymptotic formula; A062154, A062156 and A001499 are at revisions #14, #7 and #130, and say what this section reports. The A062154 formula determines (2.1) (Remark 2.1(a)), so the exact part of the report is older than “closely related” suggests; the source claims no priority for it. Its asymptotic results are not affected. The source does not cite the A283500 family or the result of Ordentlich, Parvaresh and Roth [9] for $k = n/2$ (abstract read), which does not bear on fixed k .

References

- [1] OEIS Foundation Inc., A197458, *Number of $n \times n$ binary matrices with at most two ones in each row and column*. <https://oeis.org/A197458>. Inspected record revision 25, 8 April 2020; retrieved 3 October 2026. Official data record: <https://github.com/oeis/oeisdata/blob/main/seq/A197/A197458.seq>.
- [2] R. J. Mathar, *The number of binary $n \times m$ matrices with at most k ones in each row or column*, 23 November 2014, seven pages. <https://oeis.org/A247158/a247158.pdf>.
- [3] *Filling out an $n \times n$ square grid with zeros and ones*, Math Stack Exchange question 72600 (2011), answer by joriki (2012). <https://math.stackexchange.com/questions/72600>. Original exposition of the recurrence; not a refereed publication.
- [4] OEIS Foundation Inc., A062154 and A062156. <https://oeis.org/A062154>; <https://oeis.org/A062156>. These entries cite I. P. Goulden and D. M. Jackson, *Combinatorial Enumeration*, Wiley, 1983, Problem 3.4.15. The book exercise was not directly inspected.
- [5] OEIS Foundation Inc., A001499, the exactly-two line-sum family. <https://oeis.org/A001499>. Historical attributions here are via the entry; the cited original Knuth solution was not directly inspected.
- [6] C. Greenhill, B. D. McKay, and X. Wang, *Asymptotic enumeration of sparse 0–1 matrices with irregular row and column sums*, 2006, Theorem 1.3. Author-hosted text: <https://cs.anu.edu.au/~bdm/papers/irregular.pdf>.
- [7] E. Caizergues and É. de Panafieu, *Exact enumeration of graphs and bipartite graphs with degree constraints*, EuroComb 2023, pp. 254–262, Theorem 2. <https://doi.org/10.5817/CZ.MUNI.EUROCOMB23-035>. Publisher text: <https://journals.muni.cz/eurocomb/article/view/35569>.
- [8] OEIS Foundation Inc., A283500 (*Triangle read by rows: $T(n,k)$ = number of $n \times n$ (0,1) matrices with at most k 1’s in each row or column*) and A247158. <https://oeis.org/A283500>; <https://oeis.org/A247158>. Added by the write (6 October 2026).
- [9] E. Ordentlich, F. Parvaresh, and R. M. Roth, *Asymptotic enumeration of binary matrices with bounded row and column sums*, SIAM J. Discrete Math. 26 (2012), no. 4, 1550–1575. <https://doi.org/10.1137/110857465>. Added by the write (6 October 2026); abstract read.